

Representing the equation by $V = 0$, the function V should verify the fundamental equation $d_p V \cdot d_s V + d_q V \cdot d_t V + d_r V \cdot d_u V = 0$, and we in fact have

$$\begin{aligned} d_p V &= (b^2 + c^2)p - abq - acr & + cdt - bdu, \\ d_s V &= & - cdq + bdr + (a^2 + d^2)s + abt + acu, \end{aligned}$$

and thence forming the product $d_p V \cdot d_s V$, and from it the two analogous products, we find

	$d_p V \cdot d_s V =$	$d_q V \cdot d_t V =$	$d_r V \cdot d_u V =$
p^2	0	$-abcd$	$+abcd$
q^2	$+abcd$	0	$-abcd$
r^2	$-abcd$	$+abcd$	0
s^2	$+abcd$	0	$-abcd$
t^2	$-abcd$	$+abcd$	0
u^2	0	$-abcd$	$+abcd$
pq	$-cd(b^2 + c^2)$	$+abcd$	$-cd(a^2 - b^2)$
qr	$-ad(b^2 - c^2)$	$+bd(a^2 + c^2)$	$-bd(a^2 - c^2)$
rp	$+cd(a^2 + d^2)$	$-ad(b^2 + d^2)$	$+ad(c^2 - d^2)$
st	$+bd(a^2 + d^2)$	$-ab(c^2 + d^2)$	$-ab(c^2 - d^2)$
tu	$-ad(b^2 - c^2)$	$-ad(b^2 + d^2)$	$+ad(b^2 - d^2)$
us	$+cd(a^2 + d^2)$	$-bc(a^2 - d^2)$	$-ab(c^2 - d^2)$
ps	$-a^2b^2 - c^2d^2$	$-ab(c^2 + d^2)$	$-ab(c^2 - d^2)$
qt	$+(a^2 + d^2)(b^2 + c^2)$	$+ab(c^2 + d^2)$	$-ac(c^2 + d^2)$
ru	$+a^2c^2 + b^2d^2$	$+(b^2 + d^2)(c^2 + a^2)$	$-ab(c^2 - d^2)$
pt	$+ac(b^2 + c^2)$	$+bc(c^2 + d^2)$	$+ab(c^2 + d^2)$
qu	$+ab(b^2 + c^2)$	$-b^2c^2 - a^2d^2$	$+bc(a^2 + b^2)$
rs	$-bc(b^2 + c^2)$	$-ad(b^2 - c^2)$	$+(c^2 + d^2)(d^2 - b^2)$
pu	$-bc(a^2 + d^2)$	$+bc(b^2 - d^2)$	.

by which the equation is verified.

I conclude with some remarks relating to a different part of the subject. It is shown by M. E. de Jonquières in his "Essai sur la génération des courbes géométriques, &c." *Mém. Prés. à l'Acad. de Paris*, t. XVI. (1858), that the equation $U = 0$ of any geometrical plane curve can be presented in a variety of different ways in the form

$$\begin{vmatrix} P, & Q \\ P', & Q' \end{vmatrix} = 0,$$

this being in fact the fundamental theorem of his very beautiful investigations. I am not aware that it has ever been considered whether the equation $U = 0$ of a plane curve can in general be represented in the form

$$\begin{vmatrix} P, & Q, & R \\ P', & Q', & R' \\ P'', & Q'', & R'' \end{vmatrix} = 0$$

or in the analogous forms where the left-hand side is a determinant of a higher order. As regards surfaces, the equation $U = 0$ of a geometrical surface cannot in general be presented in the form

$$\begin{vmatrix} P, & Q \\ P', & Q' \end{vmatrix} = 0,$$

nor in the similar forms where the left-hand side is a determinant of a higher order. We may consequently classify surfaces of a given order according to the forms of this nature by which they can be represented, or as I propose to term it, according to their “frangibility.” It is obvious that this question is immediately connected with that of the representation of curves in space by means of the ordinary coordinates of analytical geometry; for instance if we have a surface $U = 0$, which can be represented in the form

$$\begin{vmatrix} P, & Q, & R \\ P', & Q', & R' \\ P'', & Q'', & R'' \end{vmatrix} = 0,$$

then we can describe upon the surface curves such as

$$\begin{vmatrix} P', & Q', & R' \\ P'', & Q'', & R'' \end{vmatrix} = 0,$$

viz. the curve so represented is the curve which in conjunction with the curve ($R' = 0, R'' = 0$) makes up the complete intersection of the two surfaces $P'R'' - P''R' = 0, Q'R'' - Q''R' = 0$: the curve in question is not in general the complete intersection of any two surfaces. If the surface $U = 0$ can be represented in the form $\begin{vmatrix} P, & Q \\ P', & Q' \end{vmatrix} = 0$, then we can describe upon the surface, curves such as the curve ($P = 0, Q = 0$) which, although it is the complete intersection of two surfaces, is not the complete intersection of the given surface $U = 0$ by any other surface. But if the surface $U = 0$ cannot be represented in any such form, or as we may express it, if the surface is infrangible, then it would appear that the only curves which can be described upon the surface are those which are the complete intersection of the given surface by some other surface. The question is, I think, an interesting one in the theory of surfaces, but I doubt whether much will be done in this manner as regards the theory of curves in space, and it appears to me that there is more to be hoped for from the theory previously explained in the present paper.

2, Stone Buildings, W.C., June 2, 1859.

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ON THE SYSTEM OF CONICS HAVING DOUBLE CONTACT WITH EACH OTHER.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. III. (1860), pp. 246–250.]

CONSIDER the conics which pass through four points coinciding two and two together; the two points of each pair of coincident points are to be regarded as lying in a line which will be a tangent to the conic, and the system is thus a system of conics touching the same two lines at the same two points: or if we replace the two lines by a conic of the system, it is the system of conics having double contact with a given conic at the same two points. The lines may be spoken of as the tangents, and the points as the ineunts of the system; the line joining the two points is the axis, and the point of intersection of the two lines, the pole. It is assumed that the system contains real conics; the pole and axis must consequently be real, but the tangents and ineunts may be either all of them real or else all of them imaginary—we may for shortness say that in the former case the double contact is real, and that in the latter case it is imaginary. Consider first the case of a real double contact, the system of conics commences with an ellipse differing only slightly from the finite portion of the axis included between the ineunts, this ellipse being the first of a series of ellipses the last of which only slightly differs from a parabola, we have then a parabola, then a hyperbola differing only slightly from the parabola, and which is the first of a series of hyperbolas of which the last differs only slightly from the two tangents: all the before-mentioned conics are included within the angles at the pole or vertex of the triangle, viz. the ellipses and parabola within the angle towards the axis or base, and the hyperbolas, one branch of each of them within this angle and the other branch within the opposite angle. And it will be convenient to term these conics the “Multi- form Series.” Passing over the intermediate case, we come to a hyperbola differing only slightly from the two tangents, the first of a series of hyperbolas of which the

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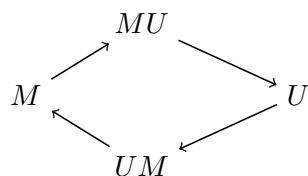
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last differs only slightly from the infinite portions, not included between the ineunts, of the axis; all these hyperbolas lying without the angles at the pole or vertex of the triangle. The last-mentioned hyperbolas may be termed the “Uniform Series.” Passing over another intermediate case we return to the multiform series, the entire system of conics forming in fact a complete cycle. The intermediate case forming the transition from the multiform to the uniform series is a conic which considered as generated by a point is the pair of tangents, but which considered as enveloped by a line is a pair of points coincident with the pole. The intermediate case forming the transition from the uniform series to the multiform series is a conic which considered as generated by a point is a pair of lines coincident with the axis, but considered as enveloped by a line is the pair of ineunts.

When the double contact is imaginary, the conics of the system may still be considered as forming a multiform and a uniform series, but the uniform series consists entirely of imaginary conics. The multiform series commences with an ellipse differing only slightly from the pole; this is the first of a series of ellipses of which the last differs only slightly from a parabola, we have next the parabola, and then a series of hyperbolas, the first of which differs only slightly from the parabola, and the last of which differs only slightly from the axis. The ellipses and parabolas lie all of them on the same side of the axis; the hyperbolas lie, one branch of each on the one side of the axis, and the other branch on the other side. The uniform series (being imaginary) of course does not admit of description. The intermediate or transition cases are the same as for a real double contact, with only the variation occasioned by the tangents and ineunts being imaginary. The conics of the entire system are considered (as in the case of a real double contact) to form a cycle, but here part of the cycle, viz. the uniform series, is imaginary.

Suppose that M and U stand for the multiform and uniform series, MU for the transition form from the multiform to the uniform series, and UM for the transition form from the uniform to the multiform series. The cycle may be represented as in the figure.



Call either of the conics MU , UM the centre of the cycle, and say that the cycle is arranged line-wise when MU is considered as the centre, and point-wise when UM is considered as the centre. We may pass from one conic of the cycle to another incentrically or excentrically, i.e. by passing towards the centre or away from the centre. The intermediates of two conics are the conics passed through in going from the first to the second incentrically, the extramediates are those passed through in going from the first to the second excentrically. In speaking of the extramediates or the intermediates of a single conic, such conic is considered as the first conic and the second conic is understood to be the centre. The nearer extramediates are those passed through previous to reaching an extremity of the cycle, the further extramediates are those passed through subsequent to reaching an extremity of the cycle. It is clear that the intermediates and the nearer extramediates make up the series multiform or uniform to which the conic belongs, and that the further extramediates are the other series; and moreover that if the cycle be arranged line-wise and point-wise successively, the intermediates of the one arrangement are the nearer extramediates of the other arrangement and vice versa; the further extramediates of each arrangement being the same.

Two points of a conic may be said to be conjunctive with respect to a given line when it is possible to pass from the one to the other along the curve without crossing the line and disjunctive in the contrary case—it being understood that both the parabola and the hyperbola are to be treated as closed curves, viz. the points at infinity of the parabola are to be considered as one and the same point, and so the points at each extremity of either asymptote of the hyperbola are to be considered as one and the same point—and in like manner two tangents of a conic are said to be conjunctive with respect to a given point, when it is possible by the revolution of the one tangent to arrive at the other tangent without sweeping through the point in question, and disjunctive in the contrary case.

Consider now any conic of the series, and call this simply the conic. Take upon the conic any two points a , a' and joining these with a variable point in the axis, we obtain the lines ab' and $a'b$ meeting the conic in the points b , b' (so that ab' and $a'b$ meet on the axis). We have then the conic in a series of points a , b , c , ... and a second series a' , b' , c' , ... which are homographically related to each other, and which possess the property that taking any two points b , c of the first series and the corresponding points b' , c' of the second series, the lines bc' , $b'c$ meet on the axis. It is immaterial how the point a is chosen, but a being chosen at pleasure, the system will obviously depend on the way in which the point a' is chosen; so that the points of a conic may be considered as homographically related to each other in an infinity of different ways. The reciprocal construction of course applies to the tangents of a conic, so that the tangents of a conic may be considered as homographically related to each other in an infinity of different ways: and not only this, but it is clear that the tangents at points homographically related to each other are also homographically related to each other, and vice versa.

The line joining corresponding points envelopes a conic, one of the conics of the system, and which may for shortness be spoken of simply as the envelope. The point of intersection of corresponding tangents generates a conic, one of the conics of a system, and which may in like manner be spoken of simply as the locus.

We may now enunciate the following theorem: Let the system be arranged line-wise, the envelope is an extramediate of the conic; viz. if a pair of corresponding points of the conic (all pairs have in this respect the same property) be conjunctive with respect to the axis, a nearer extramediate, but if disjunctive, then a further extramediate.

Let the system be arranged point-wise, the locus is an extramediate of the conic; viz. if a pair of corresponding tangents of the conic (all pairs have in this respect the same property) be conjunctive with respect to the pole, a nearer extramediate, but if disjunctive, then a further extramediate.

It is however proper to remark that when the double contact is imaginary, the pair of corresponding points or tangents (in fact any pair of points or tangents) is necessarily conjunctive, so that in this case the second alternatives in the two theorems have no application, and the envelope or locus is always a nearer extramediate. When the double contact is real, the pair of corresponding points or tangents may be either conjunctive or disjunctive, and the character (nearer or further) of the extramediate is determined accordingly.

The double contact being either real or imaginary, we may as a limiting case assume that the corresponding points or tangents coincide, the envelope or locus is in this case the conic itself. Suppose that the double contact is real, we have here two other limiting cases, viz. first, one of a pair (and therefore of each pair) of corresponding points or tangents may be situate in the axis or pass through the pole; this is the limit between the two cases of the pair of corresponding points or tangents being conjunctive and disjunctive, and therefore the envelope or locus must be the conic which is the limit between the nearer extramediates and the further extramediates, i.e. the envelope is the pair of ineunts and the locus is the pair of tangents.

Secondly, the line through a pair of corresponding points may pass through the pole, or the point of intersection of a pair of corresponding tangents may lie in the axis; the envelope or locus is here the furthest of the extramediates, viz. the envelope is the pair of points coincident with the pole, the locus is the pair of lines coincident with the axis. If considering one of the pair of corresponding points or lines as fixed, the other point or line passes through the last-mentioned limiting position, the envelope or locus returns back through the series of further extramediates.

In the statement of the preceding theorems, the envelope and locus have been considered separately, but we may if we please consider the locus as generated by the point of intersection of the tangents at the corresponding points: the locus and envelope are in this case reciprocal polars with respect to the conic; it should be noticed that the pair of corresponding points and the pair of corresponding tangents are either both conjunctive, or else both disjunctive.

2, Stone Buildings, W.C., 3rd June, 1859.

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NOTE ON THE VALUE OF CERTAIN DETERMINANTS, THE TERMS
OF WHICH ARE THE SQUARED DISTANCES OF POINTS IN
A PLANE OR IN SPACE.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. III. (1860), pp. 275–277.]

THE values of the several determinants mentioned in my paper “On a Certain Theorem in the Geometry of Position,” *Cambridge Mathematical Journal*, Old Series, t. II. (1842), p. 267, [1], are as follows:

$$(1) \begin{vmatrix} 0, & \overline{12}^2, & \overline{13}^2, & 1 \\ \overline{21}^2, & 0, & \overline{23}^2, & 1 \\ \overline{31}^2, & \overline{32}^2, & 0, & 1 \\ 1, & 1, & 1, & 0 \end{vmatrix} = \Sigma \overline{12}^2 \overline{21}^2 - \Sigma \overline{12}^2 \overline{23}^2,$$

where the Σ , Σ contain 3 and 6 terms respectively.

$$(2) \begin{vmatrix} 0, & \overline{12}^2, & \overline{13}^2, & \overline{14}^2, & 1 \\ \overline{21}^2, & 0, & \overline{23}^2, & \overline{24}^2, & 1 \\ \overline{31}^2, & \overline{32}^2, & 0, & \overline{34}^2, & 1 \\ \overline{41}^2, & \overline{42}^2, & \overline{43}^2, & 0, & 1 \\ 1, & 1, & 1, & 1, & 0 \end{vmatrix} = \Sigma \overline{12}^2 \overline{23}^2 \overline{34}^2 - \Sigma \overline{12}^2 \overline{34}^2 \overline{43}^2 - \Sigma \overline{12}^2 \overline{23}^2 \overline{31}^2,$$

where the Σ , Σ , Σ contain 24, 12 and 8 terms respectively.

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$$(3) \begin{vmatrix} 0, & \overline{12}^2, & \overline{13}^2, & \overline{14}^2, & \overline{15}^2, & 1 \\ \overline{21}^2, & 0, & \overline{23}^2, & \overline{24}^2, & \overline{25}^2, & 1 \\ \overline{31}^2, & \overline{32}^2, & 0, & \overline{34}^2, & \overline{35}^2, & 1 \\ \overline{41}^2, & \overline{42}^2, & \overline{43}^2, & 0, & \overline{45}^2, & 1 \\ \overline{51}^2, & \overline{52}^2, & \overline{53}^2, & \overline{54}^2, & 0, & 1 \\ 1, & 1, & 1, & 1, & 1, & 0 \end{vmatrix} = -\Sigma \overline{12}^2 \overline{23}^2 \overline{34}^2 \overline{45}^2 \\ - \Sigma \overline{12}^2 \overline{23}^2 \overline{34}^2 \overline{43}^2 \\ + \Sigma \overline{12}^2 \overline{23}^2 \overline{45}^2 \overline{54}^2 \\ + \Sigma \overline{12}^2 \overline{23}^2 \overline{34}^2 \overline{41}^2 \\ + \Sigma \overline{12}^2 \overline{23}^2 \overline{31}^2 \overline{45}^2$$

where the Σ , Σ , Σ , Σ , Σ contain 120, 15, 60, 30, and 40 terms respectively.

$$(4) \begin{vmatrix} 0, & \overline{12}^2, & \overline{13}^2, & \overline{14}^2 \\ \overline{21}^2, & 0, & \overline{23}^2, & \overline{24}^2 \\ \overline{31}^2, & \overline{32}^2, & 0, & \overline{34}^2 \\ \overline{41}^2, & \overline{42}^2, & \overline{43}^2, & 0 \end{vmatrix} = \Sigma \overline{12}^2 \overline{21}^2 \overline{34}^2 \overline{43}^2 - \Sigma \overline{12}^2 \overline{23}^2 \overline{34}^2 \overline{41}^2,$$

where the Σ , Σ contain 3 and 6 terms respectively.

$$(5) \begin{vmatrix} 0, & \overline{12}^2, & \overline{13}^2, & \overline{14}^2, & \overline{15}^2 \\ \overline{21}^2, & 0, & \overline{23}^2, & \overline{24}^2, & \overline{25}^2 \\ \overline{31}^2, & \overline{32}^2, & 0, & \overline{34}^2, & \overline{35}^2 \\ \overline{41}^2, & \overline{42}^2, & \overline{43}^2, & 0, & \overline{45}^2 \\ \overline{51}^2, & \overline{52}^2, & \overline{53}^2, & \overline{54}^2, & 0 \end{vmatrix} = \Sigma \overline{12}^2 \overline{23}^2 \overline{34}^2 \overline{45}^2 \overline{51}^2 \\ - \Sigma \overline{12}^2 \overline{23}^2 \overline{31}^2 \overline{45}^2 \overline{54}^2$$

where the Σ , Σ contain 24 and 20 terms respectively.

And it is proper to remark that it is not in the preceding formulæ (as in the memoir above referred to in which $\overline{12}$ denotes a distance between two points 1 and 2) assumed that $\overline{12}$ and $\overline{21}$ are equal.

The formula (1) gives, if a, b, c represent the sides, the well-known expression for the area of a triangle

$$(\text{area})^2 = \frac{1}{16}(2b^2c^2 + 2c^2a^2 + 2a^2b^2 - a^4 - b^4 - c^4).$$

Similarly the formula (2), if a, b, c, f, g, h represent the edges, viz. $\overline{23} = a, \overline{31} = b, \overline{12} = c, \overline{14} = f, \overline{24} = g, \overline{34} = h$, gives for the volume of a tetrahedron

$$\begin{aligned} (\text{volume})^2 &= \frac{1}{144} \{ b^2c^2(g^2 + h^2) + c^2a^2(h^2 + f^2) + a^2b^2(f^2 + g^2) \\ &\quad + g^2h^2(b^2 + c^2) + h^2f^2(c^2 + a^2) + f^2g^2(a^2 + b^2) \\ &\quad - a^2f^2(a^2 + f^2) - b^2g^2(b^2 + g^2) - c^2h^2(c^2 + h^2) \\ &\quad - a^2g^2h^2 - b^2h^2f^2 - c^2f^2g^2 - a^2b^2c^2 \}, \\ &= \frac{1}{144} W \quad \text{suppose.} \end{aligned}$$

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Now

$$\begin{aligned} 4 \times \text{surface} &= \sqrt{2g^2h^2 + 2h^2a^2 + 2a^2g^2 - a^4 - h^4 - g^4} \\ &\quad + \sqrt{2f^2h^2 + 2h^2b^2 + 2b^2f^2 - h^4 - b^4 - f^4} \\ &\quad + \sqrt{2g^2f^2 + 2f^2c^2 + 2c^2g^2 - g^4 - f^4 - c^4} \\ &\quad + \sqrt{2b^2c^2 + 2c^2a^2 + 2a^2b^2 - a^4 - b^4 - c^4}, \\ &= x + y + z + w \quad \text{suppose,} \end{aligned}$$

and the norm of this is

$$(x^4 + y^4 + z^4 + w^4 - 2y^2z^2 - 2z^2w^2 - 2w^2x^2 - 2x^2y^2 - 2y^2w^2 - 2x^2z^2)^2 - 64x^2y^2z^2w^2,$$

which is of the sixteenth order, and must be of the form WQ where Q is a function of a, b, c, f, g, h of the tenth order. The expression of this function is given by Prof. Sylvester in his paper "On the Relation between the Volume of a Tetrahedron &c.," *Camb. and Dubl. Math. Jour.*, t. VIII. (1853), pp. 171–178, viz. the value is

$$\begin{aligned} Q &= a^2b^2c^2\{f^4 + g^4 + h^4 + g^2h^2 + h^2f^2 + f^2g^2 + b^2c^2 + c^2a^2 + a^2b^2 - (f^2 + g^2 + h^2)(a^2 + b^2 + c^2)\} \\ &\quad + a^2g^2h^2\{f^4 + b^4 + c^4 + b^2c^2 + c^2f^2 + f^2b^2 + g^2h^2 + h^2f^2 + f^2g^2 - (f^2 + b^2 + c^2)(a^2 + g^2 + h^2)\} \\ &\quad + b^2h^2f^2\{g^4 + c^4 + a^4 + c^2a^2 + a^2g^2 + g^2c^2 + h^2f^2 + f^2h^2 + h^2g^2 - (g^2 + c^2 + a^2)(b^2 + h^2 + f^2)\} \\ &\quad + c^2f^2g^2\{h^4 + a^4 + b^4 + a^2b^2 + b^2h^2 + h^2a^2 + f^2g^2 + g^2c^2 + c^2f^2 - (h^2 + a^2 + b^2)(c^2 + f^2 + g^2)\}, \end{aligned}$$

and, as there remarked, the equation $Q = 0$ expresses the condition that the radius of the inscribed sphere may be infinite.

2, Stone Buildings, W.C., June 10th, 1859.

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NOTE ON THE EQUATION FOR THE SQUARED DIFFERENCES
OF THE ROOTS OF A CUBIC EQUATION.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. III. (1860), pp. 307–309.]

THE question of finding the equation for the squared differences of the roots, presents, in the case of a cubic equation, a peculiarity which does not occur for equations of a higher order, viz. we may in the first instance form the equation for the differences of the roots taken in a given cyclical order, and thence deduce the equation for the squared differences of the roots. Let the cubic equation be

$$U = (a, b, c, d \mid x, 1)^3 = a(x - \alpha)(x - \beta)(x - \gamma) = 0,$$

the function

$$\Pi\{\theta - (\beta - \gamma)\},$$

which equated to zero gives for θ the values $\beta - \gamma$, $\gamma - \alpha$, $\alpha - \beta$, which are the differences of the roots taken in a given circular order, has for any interchanges whatever of the roots, two values only, viz. that just written down, and the value $\Pi\{\theta + (\beta - \gamma)\}$ [which is $\Pi\{\theta - (\gamma - \beta)\}$], which may be deduced therefrom by changing first the sign of θ and then the sign of the entire expression (or what is the same thing, by changing the signs of the terms containing the even powers of θ); we may consequently write

$$\Pi\{\theta - (\beta - \gamma)\} = P - Q\sqrt{\square(\alpha, \beta, \gamma)},$$

which P , Q are symmetrical functions of the roots, and $\sqrt{\square(\alpha, \beta, \gamma)}$ or $(\alpha - \beta)(\beta - \gamma)(\gamma - \alpha)$ is a function the square of which is a symmetrical function of the roots, and such

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symmetrical functions of the roots can of course be expressed as functions of the coefficients. We have in fact

$$P = \theta^3 + \theta(2\beta\gamma - \Sigma\alpha^2) = a^{-3}\{a^3\theta^3 + 9(ac - b^2)\theta\},$$

and

$$Q = 1,$$

where $\square$ is the discriminant of the cubic function,

$$= a^2d^2 - 6abcd + 4ac^3 + 4b^3d - 3b^2c^2.$$

Consequently

$$\Pi\{\theta - (\beta - \gamma)\} = a^{-3}\{a^3\theta^3 + 9(ac - b^2)\theta - \sqrt{-27\square}\},$$

and forming the similar equation

$$\Pi\{\theta + (\beta - \gamma)\} = a^{-3}\{a^3\theta^3 + 9(ac - b^2)\theta + \sqrt{-27\square}\},$$

multiplying the two equations together and writing u in the place of θ^2 , we find

$$\Pi\{u - (\beta - \gamma)^2\} = a^{-6}\{[a^3u + 9(ac - b^2)]^2u + 27\square\},$$

and the equation for the squared differences of the roots is thus seen to be

$$\{a^3u + 9(ac - b^2)\}^2u + 27\square = 0,$$

or what is the same thing

$$a^6u^3 + 18a^3(ac - b^2)u^2 + 81(ac - b^2)^2u + 27\square = 0.$$

I remark that if ω is an imaginary cube root of unity (so that $(\omega - \omega^2)^2 = -3$, $\omega - \omega^2$ being thus only another form of $\sqrt{-3}$) then if in the expression for $\Pi\{\theta - (\beta - \gamma)\}$ we write $\frac{\theta}{(\omega - \omega^2)a}$ in the place of θ , the equation assumes the more simple form

$$\Pi\{\theta - \frac{1}{3}a(\omega - \omega^2)(\beta - \gamma)\} = \theta^3 - 3(ac - b^2)\theta - a^3\sqrt{\square},$$

which if U be the cubic function, H the Hessian $= (ac - b^2, ad - bc, bd - c^2 \mid x, y)^2$, and $\square$ the discriminant as before, is a particular case (obtained by writing $x = 1, y = 0$) of the equation

$$\Pi\{\theta - \frac{1}{3}a(\omega - \omega^2)(x - \alpha y)\} = \theta^3 - 3H\theta - U\sqrt{\square},$$

which equation can be at once obtained from the equation (where Φ is the cubi covariant of the cubic function)

$$\sqrt[3]{\frac{1}{2}(\Phi + U^2\sqrt{\square})} - \sqrt[3]{\frac{1}{2}(\Phi - U^2\sqrt{\square})} = \frac{1}{3}a(\omega - \omega^2)(\beta - \gamma)(x - \alpha y),$$

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given in my Fifth Memoir on Quantics, *Phil. Trans.*, t. CXLVIII. (1858), [156]. For writing for a moment

$$\theta = \sqrt[3]{X} - \sqrt[3]{Y},$$

we find

$$\theta^3 = X - Y - 3\sqrt[3]{XY}\theta,$$

or

$$\theta^3 + 3\sqrt[3]{XY}\theta - (X - Y) = 0,$$

where $\sqrt[3]{XY} = \sqrt[3]{\frac{1}{4}(\Phi^2 - U^4\square)}$, which by the equation

$$\Phi^2 - U^4\square = -4H^3$$

(given in the Memoir) is $= -H$, and $(X - Y)$ is $= U^2\sqrt{\square}$, so that the equation in θ is, as above, $\theta^3 - 3H\theta - U^2\sqrt{\square} = 0$, an equation which is satisfied by $\theta = \frac{1}{3}a(\omega - \omega^2)(\beta - \gamma)(x - \alpha y)$; and the other two roots being of course of the like form, the cubic function in θ is equal to $\Pi\{\theta - \frac{1}{3}a(\omega - \omega^2)(\beta - \gamma)(x - \alpha y)\}$ which proves the theorem.

2, Stone Buildings, W.C., Nov. 3rd, 1859.

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NOTE ON THE CURVATURE OF A PLANE CURVE AT A DOUBLE POINT, AND ON THE CURVATURE OF SURFACES.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. III. (1860), pp. 322–326.]

THE radius of curvature of a plane curve at a double point is most readily determined as follows: viz. if x, y be the coordinates of the double point, θ the inclination to the axis of x of the tangent to the branch under consideration, and u, v the coordinates measured from the double point as origin, parallel and perpendicular to the tangent, of the consecutive point on such branch, then the radius of curvature for the branch in question is

$$\frac{u^2}{2v};$$

the coordinates of the consecutive point are

$$x + u \cos \theta - v \sin \theta,$$

$$y + u \sin \theta + v \cos \theta,$$

and the value of $u^2 \div 2v$ is to be found by substituting these expressions for x, y in the equation of the curve. Let $U = 0$ be the equation of the curve; then at the double point, U and the differential coefficients of the first order vanish, let those of the second order be (a, b, c) and those of the third order $(\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{d})$, then substituting we have

$$\begin{aligned} & \frac{1}{2}(a, b, c)(u \cos \theta - v \sin \theta, u \sin \theta + v \cos \theta)^2 \\ & + \frac{1}{6}(\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{d})(u \cos \theta - v \sin \theta, u \sin \theta + v \cos \theta)^3 = 0. \end{aligned}$$

The coefficient of u^2 is

$$\frac{1}{2}(a, b, c)(\cos \theta, \sin \theta)^2,$$

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which vanishes, since it is by putting this function equal to zero that we find the direction of the tangents at the double point. For the present purpose we are concerned with one of the two branches only, and in all that follows the ratio $\cos \theta : \sin \theta$ will denote a determinate root of the quadratic equation; viz. the root which corresponds to the branch in question.

The equation takes therefore the form

$$Buv + \frac{1}{2}Cv^2 + \frac{1}{6}Du^3 + \&c. = 0,$$

which might be satisfied by assuming $Buv + \frac{1}{2}Cv^2 = 0$, but the values so obtained belong to the branch which does not touch the tangent; the proper solution is

$$Buv + \frac{1}{6}Du^3 = 0,$$

and thence

$$\frac{u^2}{2v} = -\frac{3B}{D},$$

or substituting for B, D their values, the radius of curvature is

$$= -\frac{3(a, b, c)(\cos \theta, \sin \theta)(-\sin \theta, \cos \theta)}{(\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{d})(\cos \theta, \sin \theta)^3},$$

where the expression in the numerator is

$$= (-a + c) \sin \theta \cos \theta + b(\cos^2 \theta - \sin^2 \theta),$$

$$= \frac{1}{2}[(c - a) \sin 2\theta + 2b \cos 2\theta].$$

The curvature, at the point of contact, of the curve in which any surface is intersected by a tangent plane cannot be found by the ordinary theory of the curvature of surfaces, being (by reason that the point of contact is a double point on the curve) a case of exception from that theory. The foregoing method may be applied to the solution of the question as follows: Let $U = 0$ be the equation of the surface, and suppose that (x, y, z) refer to the point of contact of the tangent plane. Let L, M, N be the first differential coefficients of U at this point, (a, b, c, f, g, h) the second differential coefficients, $(\alpha, \beta, \gamma, \delta, \epsilon, \zeta, \eta, \theta, \iota)$ the third differential coefficients. Let λ, μ, ν be proportional to the cosines of the inclinations to the axis of the tangent to one of the two branches through the double point; the ratios $\lambda : \mu : \nu$ are determined by

$$(L, M, N)(\lambda, \mu, \nu) = 0,$$

$$(a, b, c, f, g, h)(\lambda, \mu, \nu)^2 = 0.$$

Let u, v be proportional to the coordinates of a consecutive point measured, from the point of contact as origin, in the direction of the tangent and in the perpendicular direction in the tangent plane. The cosines of the inclinations of u to the axes are as $\lambda : \mu : \nu$; those for the inclinations of the normal to the axes are as $L : M : N$; hence for the coordinate v which is perpendicular to the plane of the last-mentioned

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two lines, the cosines of the inclinations to the axes are as $N\mu - M\nu : L\nu - N\lambda : M\lambda - L\mu$, and the coordinates of the consecutive point may be taken to be

$$x + \lambda u + (N\mu - M\nu)v,$$

$$y + \mu u + (L\nu - N\lambda)v,$$

$$z + \nu u + (M\lambda - L\mu)v.$$

Substituting these values in the equation of the surface, the terms involving the first powers of u, v vanish, and the term involving u^2 also vanishes in virtue of the relation $(a, b, c, f, g, h)(\lambda, \mu, \nu)^2 = 0$. The equation consequently becomes

$$Buv + \frac{1}{2}Cv^2 + \frac{1}{6}Du^3 + \dots = 0,$$

and we have as before for the branch in question,

$$\frac{u^2}{2v} = -\frac{3B}{D}.$$

In the present case u, v have been taken, not as before equal, but only proportional, to the coordinates of the consecutive point measured from the point of contact parallel and perpendicular to the tangent, the values of the coordinates are in fact

$$\sqrt{\lambda^2 + \mu^2 + \nu^2} u, \quad \sqrt{L^2 + M^2 + N^2} \sqrt{\lambda^2 + \mu^2 + \nu^2} v,$$

and the expression for the radius of curvature is

$$= \frac{u^2}{2v} \cdot \frac{\sqrt{\lambda^2 + \mu^2 + \nu^2}}{\sqrt{L^2 + M^2 + N^2} \sqrt{\lambda^2 + \mu^2 + \nu^2}} \cdot \sqrt{\lambda^2 + \mu^2 + \nu^2},$$

or substituting for $\frac{u^2}{2v}$ the value $-\frac{3B}{D}$ and for B, D their values, the radius of curvature is

$$= -\frac{3\sqrt{\lambda^2 + \mu^2 + \nu^2}}{\sqrt{L^2 + M^2 + N^2}} \cdot \frac{(a, b, c, f, g, h)(\lambda, \mu, \nu)(N\mu - M\nu, L\nu - N\lambda, M\lambda - L\mu)}{(\alpha, \beta, \gamma, \delta, \epsilon, \zeta, \eta, \theta, \iota)(\lambda, \mu, \nu)^3},$$

where, as already noticed, the ratios $\lambda : \mu : \nu$ are determined by the equations

$$(L, M, N)(\lambda, \mu, \nu) = 0,$$

$$(a, b, c, f, g, h)(\lambda, \mu, \nu)^2 = 0,$$

the system of roots selected being that which corresponds to the branch under consideration. It may be noticed that these two equations give

$$(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H})(L, M, N)^2 \cdot (a, b, c, f, g, h)(\lambda, \mu, \nu)^2 - K \cdot [(L, M, N)(\lambda, \mu, \nu)]^2 = 0,$$

where as usual

$$(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H}) = (bc - f^2, ca - g^2, ab - h^2, gh - af, hf - bg, fg - ch),$$

$$K = abc - af^2 - bg^2 - ch^2 + 2fgh,$$

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and the expression on the left-hand side considered as a function of (λ, μ, ν) is decomposable into a pair of factors. Selecting the proper factor and equating it to zero, we have in addition to the linear equation

$$(L, M, N)(\lambda, \mu, \nu) = 0,$$

a new linear equation; these two equations determine the ratios $\lambda : \mu : \nu$.

P.S.—I remark that the process adopted for finding the radius of curvature at a double point of a plane curve is the most simple one for the case of an ordinary point on the curve. In fact let $U = 0$ be the curve, (x, y) the coordinates of the point in question, L, M the corresponding values of the first differential coefficients, (a, b, c) those of the second differential coefficients. The coordinates of the consecutive point are

$$x + u \cos \theta - v \sin \theta,$$

$$y + u \sin \theta + v \cos \theta.$$

Substituting these in the equation of the curve, we have

$$L(u \cos \theta - v \sin \theta) + M(u \sin \theta + v \cos \theta) + \frac{1}{2}(a, b, c)(u \cos \theta - v \sin \theta, u \sin \theta + v \cos \theta)^2 = 0,$$

the coefficient of u must be zero, or we have

$$L \cos \theta + M \sin \theta = 0,$$

giving

$$\sin \theta = \frac{L}{\sqrt{L^2 + M^2}}, \quad \cos \theta = -\frac{M}{\sqrt{L^2 + M^2}},$$

and the equation may be reduced to

$$\sqrt{L^2 + M^2} v + \frac{1}{2}(a, b, c)(\cos \theta, \sin \theta)^2 u^2 = 0,$$

or what is the same thing

$$(L^2 + M^2)^{\frac{3}{2}} v + \frac{1}{2}(a, b, c)(M, -L)^2 u^2 = 0,$$

whence the radius of curvature

$$= -\frac{u^2}{2v} = \frac{(L^2 + M^2)^{\frac{3}{2}}}{(a, b, c)(M, -L)^2},$$

which is the ordinary form for the radius of curvature at any point of a curve represented by the equation $U = 0$.

2, *Stone Buildings, W.C., October 2nd, 1859.*

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ON SOME NUMERICAL EXPANSIONS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. III. (1860), pp. 366–369.]

THERE are in the theory of series and the calculus of finite differences and the differential calculus, various sets of numbers, such for instance as Bernoulli's Numbers, or the successive differences of 0^m , which from their more or less frequent occurrence and the complexity of their law it is desirable to tabulate. To these belong, I think, the coefficients of the successive powers of x in the expansion of a given power of $\frac{\log(1+x)}{x}$, or when the index of the power of the function is an indeterminate quantity r , then the coefficients of the several terms of the rational and integral functions of r which form the coefficients of the successive powers of x in the expansion in question. We have

$$\log \frac{\log(1+x)}{x} = -\frac{1}{2}x + \frac{5}{24}x^2 - \frac{1}{8}x^3 + \frac{251}{2880}x^4 - \frac{19}{288}x^5 + \frac{19087}{362880}x^6 - \&c.,$$

which is most easily verified by differentiating and multiplying by $\log(1+x)$, which gives on the left-hand side $\frac{1}{1+x}$, and on the right-hand side the expansion $1 - x + x^2 - \&c$. And from the above, multiplying each side by r , and taking the exponential, we find

$$\begin{aligned} \left\{ \frac{\log(1+x)}{x} \right\}^r &= 1 - \left(\frac{r}{2}\right)x \\ &\quad + (A_2r^2 + A'_2r)x^2 \\ &\quad - (A_3r^3 + A'_3r^2 + A''_3r)x^3 \\ &\quad + \&c. \end{aligned}$$

which may be verified by writing $r = 1$.

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As an instance of the use of the preceding table, let it be required to find expressions for the combinations without repetitions of the series of natural numbers $1, 2, 3, \dots (k-1)$, or what is the same thing for the coefficients of the powers of k in $k(k-1)(k-2)\dots(k-n+1)$. We have

$$k(k-1)(k-2)\dots(k-n+1) = A_1^n k^n - A_2^n k^{n-1} + A_3^n k^{n-2} - \&c.$$

whence

$$\begin{aligned} &= \Pi n \cdot \text{coefficient } x^n \text{ in } (1+x)^k; \\ (-)^r A_r^n &= \Pi n \cdot \text{coefficient } x^n k^{n-r} \text{ in } (1+x)^k \\ &= \Pi n \cdot \text{coefficient } x^n k^{n-r} \text{ in } e^{k \log(1+x)} \\ &= \frac{\Pi n}{\Pi(n-r)} \cdot \text{coefficient } x^n \text{ in } \{\log(1+x)\}^{n-r} \\ &= \frac{\Pi n}{\Pi(n-r)} r \cdot \text{coefficient of } x^r \text{ in } \left\{ \frac{\log(1+x)}{x} \right\}^{n-r}. \end{aligned}$$

Thus

$$\begin{aligned} A_2^n &= n\{\tfrac{1}{2}(n-1)\}, \\ A_3^n &= n(n-1)\{\tfrac{1}{6}(n-2)^2 + \tfrac{1}{4}(n-2)\}, \\ A_4^n &= n(n-1)(n-2)\{\tfrac{1}{24}(n-3)^3 + \tfrac{1}{8}(n-3)^2 + \tfrac{1}{8}(n-3)\}, \end{aligned}$$

and so on, as far as the expansion has been effected. It may be remarked that the general expression for the algebraical transcendent A_r^n is given in Dr Schläfli's paper, "Sur les coefficients du développement du produit $(1+x)(1+2x)\dots[1+(n-1)x]$ suivant les puissances ascendantes de x ." *Crelle*, t. XLIII. [1852], pp. 1–22, but the law is a very complicated one.

We have

$$\log(1+x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4 + \frac{1}{5}x^5 - \frac{1}{6}x^6 + \frac{1}{7}x^7 - \&c.$$

and dividing by x , and taking the logarithm, we have as before

$$\log\left(\frac{1}{x}\log(1+x)\right) = -\frac{1}{2}x + \frac{5}{24}x^2 - \frac{1}{8}x^3 + \frac{251}{2880}x^4 - \frac{19}{288}x^5 + \frac{19087}{362880}x^6 - \&c.,$$

which may be considered as the first of a series of logarithmic derivatives, viz., dividing by $-\frac{1}{2}x$ and taking the logarithm we have

$$\log\left(-\frac{2}{x}\log\left(\frac{1}{x}\log(1+x)\right)\right) = -\frac{5}{12}x + \frac{49}{720}x^2 - \frac{959}{30240}x^3 + \frac{6515}{169344}x^4 - \&c.,$$

and by the like process

$$\log\left(-\frac{12}{5x}\log\left(-\frac{2}{x}\log\left(\frac{1}{x}\log(1+x)\right)\right)\right) = -\frac{49}{120}x + \frac{34471}{604800}x^2 - \frac{2787221}{101606400}x^3 + \&c.,$$

and so on.

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Suppose in general that

$$\phi x = x + B_1x^2 + C_1x^3 + \dots,$$

and let it be required to find the r^{th} function $\phi^r x$. It is easy to see that the successive coefficients are rational and integral functions of r of the degrees 1, 2, 3, &c. respectively; we have in fact

$$\begin{aligned}\phi^r x &= \phi^0 x \\ &+ \frac{r}{1} (\phi^1 x - \phi^0 x) \\ &+ \frac{r \cdot r-1}{1 \cdot 2} (\phi^2 x - 2\phi^1 x + \phi^0 x) \\ &+ \frac{r \cdot r-1 \cdot r-2}{1 \cdot 2 \cdot 3} (\phi^3 x - 3\phi^2 x + 3\phi^1 x - \phi^0 x), \\ &+ \&c.,\end{aligned}$$

and by successive substitutions,

$$\begin{aligned}\phi^0 x &= x, \\ \phi^1 x &= x + B_1x^2 + C_1x^3 + D_1x^4 + \dots, \\ \phi^2 x &= x + 2B_1x^2 + (2B_1^2 + 2C_1)x^3 + (B_1^3 + 5B_1C_1 + 2D_1)x^4 + \dots, \\ \phi^3 x &= x + 3B_1x^2 + (6B_1^2 + 3C_1)x^3 + (9B_1^3 + 15B_1C_1 + 3D_1)x^4 + \dots.\end{aligned}$$

Whence

$$\begin{aligned}\phi^r x &= x \\ &+ rB_1x^2 \\ &+ \{(r^2 - r)B_1^2 + rC_1\}x^3 \\ &+ \{(\frac{1}{6}r^3 - \frac{1}{2}r^2 + \frac{1}{3}r)B_1^3 + (\frac{5}{2}r^2 - \frac{5}{2}r)B_1C_1 + rD_1\}x^4 \\ &+ \&c.\end{aligned}$$

It would, I think, be worth while to continue the expansion some steps further.

2, Stone Buildings, W.C., Oct. 2nd, 1859.

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A DISCUSSION OF THE STURMIAN CONSTANTS FOR CUBIC
AND QUARTIC EQUATIONS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. IV. (1861), pp. 7–12.]

FOR the cubic equation

$$(a, b, c, d \mid x, 1)^3 = 0,$$

the Sturmiian Constants (or leading coefficients of the Sturmiian functions) are

$$a, \quad a, \quad b^2 - ac, \quad -a^2d^2 + 6abcd - 4ac^3 - 4b^3d + 3b^2c^2.$$

If the signs of the constants, that is, of the functions for $+\infty$, are	then the signs of the functions for $-\infty$ are	
$+++$	$-+-$	three real roots. case cannot occur. one real root.
$+++$	$-+++$	
$++-$	$---$	
$+++-$	$--+-$	
$++--$	$--++$	

The second case would give a loss of variations of sign in passing from $+\infty$ to $-\infty$, which is inconsistent with Sturm's theorem. To show à posteriori that the case cannot occur, we may form the identical equation

$$(a^2d - 3abc + 2b^3)^2 = -a^2(-a^2d^2 + 6abcd - 4ac^3 - 4b^3d + 3b^2c^2) + 4(b^2 - ac)^3,$$

and, this being so, then in the case in question, the right-hand side would consist of two terms, each of them negative, while the left-hand side is essentially positive.

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In the particular case where the third constant vanishes, or $b^2 - ac = 0$, we have

$$\begin{aligned} -a^2d^2 + 6abcd - 4ac^3 - 4b^3d + 3b^2c^2 &= -(ad - bc)^2 + 4(b^2 - ac)(c^2 - bd) \\ &= -(ad - bc)^2, \text{ is negative;} \end{aligned}$$

hence, regarding the evanescent term as being at pleasure positive or negative, we have in each case a combination of signs corresponding to one real root.

The general result (which is well known) is, that there are three real roots or one real root according as

$$-a^2d^2 + 6abcd - 4ac^3 - 4b^3d + 3b^2c^2$$

is positive or negative.

For the quartic equation $(a, b, c, d, e \mid x, 1)^4 = 0$, the Sturmiian constants are

$$a, \quad a, \quad b^2 - ac, \quad 3aJ + 2(b^2 - ac)I, \quad I^3 - 27J^2,$$

if, as usual,

$$\begin{aligned} I &= ae - 4bd + 3c^2, \\ J &= ace - ad^2 - b^2e + 2bcd - c^3. \end{aligned}$$

If signs of constants, i.e. of functions for $+\infty$, are	then signs of functions for $-\infty$ are	
$+++++$	$+- -++$	4 real roots. no real root.
$++-++$	$+- +++$	
$+++ -+$	$+++ -+$	
$++-++$		
$++++-$	$+- +--$	2 real roots. 2, cannot occur. 2 real roots.
$++-+-$	$+++ -$	
$+++ -+$	$+- +- -$	
$++- --$		

The non-existing combination of signs is

$$\begin{aligned} I^3 - 27J^2 &= -, \\ 3aJ + 2(b^2 - ac)I &= +, \\ b^2 - ac &= -. \end{aligned}$$

To show à posteriori that this case cannot occur, write

$$\Psi = a^2d - 3abc + 2b^3,$$

$$X = 3aJ + 2(b^2 - ac)I,$$

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then we have identically

$$9(3a^2J^2 + X^2)\Psi^2 = -4a^2X^3 + 36(b^2 - ac)^2X^2 - 4a^2(b^2 - ac)^3(I^3 - 27J^2),$$

which is impossible under the given combination of signs, since the left-hand side would be positive, and the right-hand side negative.

To prove the above identity—the relation $JU^4 - IU^2H + 4H^3 + \Phi^2 = 0$, between the covariants of the quartic, gives

$$a^2J + a^2(b^2 - ac)I - 4(b^2 - ac)^3 + \Phi^2 = 0,$$

or, what is the same thing,

$$\Phi^2 = -a^2J - a^2(b^2 - ac)I + 4(b^2 - ac)^3.$$

But

$$X = 3aJ + 2(b^2 - ac)I,$$

and thence

$$3\Phi^2 + a^2X = -a^2(b^2 - ac)I + 12(b^2 - ac)^3,$$

$$3\Phi^2 = -a^2X - a^2(b^2 - ac)I + 12(b^2 - ac)^3,$$

and the identity will be true, if

$$\begin{aligned} (3X^2 + 9a^2J^2) \left[-X - (b^2 - ac)I + 12\frac{(b^2 - ac)^3}{a^2} \right] \\ = -4X^3 + 36\frac{(b^2 - ac)^2}{a^2}X^2 - 4(b^2 - ac)^3(I^3 - 27J^2). \end{aligned}$$

This gives

$$(3X^2 + 9a^2J^2)[-X - (b^2 - ac)I] = -4X^3 - 4(b^2 - ac)^3I^3,$$

or, what is the same thing,

$$(3X^2 + 9a^2J^2)\{X + (b^2 - ac)I\} = 4\{X^3 + (b^2 - ac)^3I^3\},$$

or, dividing by $X + (b^2 - ac)I$,

$$3X^2 + 9a^2J^2 = 4\{X^2 - X(b^2 - ac)I + (b^2 - ac)^2I^2\},$$

and reducing

$$X^2 - 4X(b^2 - ac)I - 9a^2J^2 + 4(b^2 - ac)^2I^2 = 0,$$

or finally

$$\{X - 3aJ - 2(b^2 - ac)I\}\{X + 3aJ - 2(b^2 - ac)I\} = 0,$$

which is true in virtue of

$$X = 3aJ + 2(b^2 - ac)I,$$

and the identity is thus proved.

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The general conclusion is,

if $I^3 - 27J^2$ is positive, the four roots are all real or all imaginary,

viz., all real if $b^2 - ac$ and $3aJ + 2(b^2 - ac)I$ are both positive, imaginary if otherwise. But if $I^3 - 27J^2$ is negative, the roots are two of them real, and the other two imaginary.

The following special cases may be noticed,

1°. $b^2 - ac = 0$, here

$$9(3a^2J^2 + X^2)\Psi^2 = -4a^2X^3, \text{ or } X = 3aJ + 2(b^2 - ac)I = 3aJ, \text{ is negative,}$$

so that,

if $I^3 - 27J^2$ is +, the roots are all imaginary;

if $I^3 - 27J^2$ is -, the roots are two real and two imaginary.

2°. $X = 3aJ + 2(b^2 - ac)I = 0$, here

$$27a^2J^2\Psi^2 = -4a^2(b^2 - ac)^3(I^3 - 27J^2),$$

or $b^2 - ac$, $I^3 - 27J^2$ are of opposite signs, and if

$b^2 - ac = -$, $I^3 - 27J^2 = +$, the roots are all imaginary,

$b^2 - ac = +$, $I^3 - 27J^2 = -$, the roots are two real and two imaginary.

3°. $b^2 - ac = 0$, $X = 3aJ + 2(b^2 - ac)I = 0$, here $J = 0$, that is,

$$2bcd - ad^2 - c^3 = 0, \text{ or } (ad - bc)^2 + c^2(ac - b^2) = 0, \text{ or } ad - bc = 0, \text{ and } I^3 - 27J^2 = I^3,$$

$$I = ae - 4bd + 3c^2 = ae - 4\frac{b^4}{a^2} + 3\frac{b^4}{a^2} = ae - \frac{b^4}{a^2} = \frac{1}{a^2}(a^3e - b^4),$$

whence

$I = +$, the roots are all imaginary.

$I = -$, the roots are two real and two imaginary.

This is easily verified, in fact $ac - b^2 = 0$, $ad - bc = 0$, give $c = \frac{b^2}{a}$, $d = \frac{bc}{a} = \frac{b^3}{a^2}$, and the equation becomes

$$ax^4 + 4bx^3 + 6\frac{b^2}{a}x^2 + 4\frac{b^3}{a^2}x + e = 0,$$

or, which is the same thing,

$$\frac{1}{a^3}\{(ax + b)^4 + (a^3e - b^4)\} = 0,$$

so that the roots are all imaginary, or two real and two imaginary, according to the sign of $a^3e - b^4$ as above.

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if the Sturmian Constants are

$$a, \quad a, \quad C, \quad D, \quad E, \quad F,$$

 C, D, E, F
$$\left. \begin{array}{l} - + + + \\ + - + + \\ + + - + \\ + + + - \end{array} \right\} 1 \text{ real root, 4 imaginary roots.}$$
$$\left. \begin{array}{l} + + - - \\ + - - - \\ - + + - \\ - - + - \end{array} \right\} \text{3 real roots, 2 imaginary roots.}$$
$$\left. \begin{array}{l} - + + - \\ + - - + \\ - + - \end{array} \right\} \text{cases which do not occur.}$$

2, *Stone Buildings, W.C., September 29, 1859.*

291.

ON THE DEMONSTRATION OF A THEOREM RELATING TO THE
MOMENTS OF INERTIA OF A SOLID BODY.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. IV. (1861), pp. 25–27.]

CONSIDERING in the first instance the analogous question in plano, let $a = \int x^2 dm$, $b = \int y^2 dm$, $h = \int xy dm$, where the integration extends over any closed figure whatever, then it is to be shown that the equations

$$(a, h, b \mid p, q)^2 = 1, \quad \text{and} \quad (a + b)(p^2 + q^2) - (a, h, b \mid p, q)^2 = 1,$$

represent respectively ellipses.

If in the first case, by a transformation of coordinates,

$$(a, h, b \mid p, q)^2 = a_1 p_1^2 + b_1 q_1^2,$$

then a_1, b_1 are the roots of the quadratic equations,

$$\begin{vmatrix} a - \rho, & h \\ h, & b - \rho \end{vmatrix} = 0,$$

and if in the second case,

$$(a + b)(p^2 + q^2) - (a, h, b \mid p, q)^2 = a_1 p_1^2 + b_1 q_1^2,$$

then a_1, b_1 are the roots of

$$\begin{vmatrix} b - \rho, & -h \\ -h, & a - \rho \end{vmatrix} = 0,$$

the two equations being in fact the same equation,

$$\rho^2 - (a + b)\rho + ab - h^2 = 0,$$